

EDUCATION

ACADEMIC HIGHLIGHTS

Exchange Student, Dept. of Information and Communication Engineering

The University of Tokyo

April 2027 – July 2027 (Expected) Tokyo, Japan

B.S. in Computer Science

National Yang Ming Chiao Tung University (NYCU)

Sept. 2023 – June 2027 (Expected) Hsinchu, Taiwan

- GPA: 4.12/4.3; department rank: Top 15.23% (29/197).
- Enrolled in NYCU's Art and Music Cross-Domain Program as a second specialization.
- Selected coursework (all A+): Algorithms; Data Structures & Object-Oriented Programming (Top 5%); Machine Learning; Linear Algebra; Computer System Administration; and graduate-level Cryptographic Engineering.

HONORS & AWARDS

Academic Excellence Award

National Yang Ming Chiao Tung University

Fall 2023 Hsinchu, Taiwan

- Ranked 2nd in the department (Top 1.05%) with a semester GPA of 4.27/4.3.

Scholarship for Excellence in Fundamental Courses

National Yang Ming Chiao Tung University

Spring 2024 Hsinchu, Taiwan

- Recognized for ranking in the top 5% of Data Structures and Object-Oriented Programming.

PUBLICATION

Bring Music The Horizon:
Music-Driven 360° Video Generation

Kai Hsu Tsai, Yong Wei Fu, Hung I Yang, Yu-Chih Chen

2026 arXiv GitHub AVGenL, ECCV 2026 Project

- Accepted by the Third Workshop on Audio-Visual Generation and Learning (AVGenL); presented as both a poster and a short oral talk at ECCV 2026.
- Undergraduate Research Assistant in the M³ Lab under Prof. Yu-Chih Chen, with NSTC Undergraduate Research Project funding (Summer 2025–Present).
- Engineered a Music Information Retrieval pipeline that segments songs at structural downbeats and predicts continuous valence–arousal trajectories for visual guidance.
- Retrained EmotiCrafter on a self-curated dataset to reduce human-activity artifacts and extract cleaner emotional guidance vectors.
- Integrated SEGA guidance with image-to-video models to synthesize temporally and emotionally consistent 360° transition videos.

GPA: 4.12/4.3

Department Rank: 29/197 (Top 15.23%)

General Programming Examination: 560/600

RESEARCH INTERESTS

- Multimodal AI
- Audio-Visual Generation
- Music Information Retrieval
- Computer Vision
- Human-Centered AI

TECHNICAL SKILLS

- C/C++
- Python
- Machine Learning
- Computer Vision
- Generative AI
- Linux / UNIX
- Git

LANGUAGES

English

TOEFL iBT 99 (CEFR C1)

Dec. 6, 2025

R29 / L23 / S24 / W23

Japanese

JLPT N1 – 157/180

July 2, 2023

95.6th percentile

CEFR C1 reference

Mandarin Chinese / Taiwanese

Native

SELECTED PROJECTS & COMPETITIONS

CAT: Context-Aware Bot

2nd Place Overall | 1st Place, AMD Track

Sept. 19–20, 2026 Meichu Hackathon

- Built a Physical AI companion combining an ESP32-based plush robot with pressure, microphone, camera, and LCD-face modules.
- Deployed the speech-to-text model and contributed to the dataflow across the robot, an AMD local AI PC, and a remote AMD Instinct MI300 server.
- Proposed the structured fields for 30-second user-activity snapshots, enabling efficient context extraction and persistent working-session memory for context-aware emotional support.

LINE ツォンサー: Capturing Everyday Happiness

Semifinalist, LINE FRESH 2025 Campus Competition

Sept.–Nov. 2025 Team Project

- Designed and prototyped a low-friction LINE Bot for recording text and image-based life events, generating AI-assisted daily, weekly, and monthly reflections, searching memories semantically, and tracking personal growth goals.
- Validated the concept through 322 survey responses and 10 user interviews; 82% of respondents believed frictionless recording could improve well-being.
- Iterated the prototype from user feedback, replacing conventional to-do management with low-pressure goal and growth tracking and adding customizable reminders and richer visual recaps.

Customized Dressing Benchmark (C.D.B.)

AI-Driven Fashion Advisor

May 2025 NYCU GitHub

- Built a fashion-advice system using Fashion-CLIP for visual features and LLaVA to generate personalized outfit recommendations.
- Compared Fashion-CLIP embeddings against SimCLR and ResNet baselines to evaluate representation quality.
- Received the Most Popular Award and Most Creative Presentation Award among 70 teams.

Four-Player Online Mahjong Simulator

C / TCP Client--Server Application

Sept. 2024–Jan. 2025 NYCU GitHub

- Co-developed a four-player online Mahjong game in C using a TCP client–server architecture, synchronizing player connections, turns, and shared game state across four clients.
- Designed a custom communication protocol and server-side game engine to coordinate player actions, handle disconnections, and manage multi-round sessions.
- Implemented the game’s rule engine, including tile distribution, turn progression, competing-action resolution, special-tile handling, and recursive winning-hand detection, alongside an interactive terminal interface.

CREATIVE & INTERDISCIPLINARY WORK

5 Centimeters per Second — Film Re-Sound

Exhibited Work, 2026 創・藝百匯 Joint Exhibition

📅 Mar. 2026 📍 NYCU

- Re-sounded a seven-minute film excerpt by removing its original soundtrack and reconstructing Japanese dialogue, ambience, train and station audio, and Foley.
- Recorded and synchronized Japanese dialogue and newly produced or sourced sounds to rebuild the scene’s pacing and atmosphere.
- Performed as the drummer of the student jazz ensemble at the exhibition’s opening reception.

JAPANESE LANGUAGE & CROSS-CULTURAL ENGAGEMENT

Japanese Student Counselor & Guide

NYCU Taiwan Culture & Tech Summer School

📅 Summers 2024 & 2025 📍 Hsinchu, Taiwan

- Selected as program staff for two consecutive years to support cultural exchange between Taiwanese and Japanese students.
- Systematically identified operational and program-design issues across logistics, accommodation, communication, curriculum accessibility, and site visits; volunteered as the sole student counselor at the post-program review and raised these issues directly with university program leadership.
- Interpreted technical topics, including semiconductor physics, from Mandarin to Japanese; coordinated tours and supported participant communication and safety.

Special Award, 10th AGC Japanese Presentation Contest

AGC Display Glass Taiwan

📅 Dec. 2023 📍 Taichung, Taiwan

- Presented in Japanese an AI-driven proposal to support people experiencing depression.